SHARED MEMORY IN IPC

BY → VARUN KUMAR R

INTRODUCTION

IPC enables communication between processes, allowing them to share data and coordinate activities efficiently.

Two main types of IPC exist: Shared Memory for direct data access, and Message Passing for structured communication.

SHARED MEMORY

Shared Memory is a communication method where multiple processes share a common memory area to exchange data

example :

- A whiteboard is shared space
- Teacher writes → Students read
- Students can also write → others see

Here

- Whiteboard = Shared Memory
- People = Processes

- Process 1 → one running program / writes data into shared memory
- Process 2 → another running program / reads data from shared memory
- Shared Memory Area → a common space between them
- Both processes use the same memory to communicate

- they are two mechanisms (methods) used to implement Inter Process Communication (IPC)

1.Pipes

- A Pipe is a section of shared memory used by processes for communication.

It follows a client-server architecture

- Process that creates the pipe → Pipe Server
- Process that connects to the pipe → Pipe Client

Types of Pipes

Anonymous Pipes

- No name
- Unidirectional only
- Used between two processes

Named Pipes

- Has a name
- Can be unidirectional or bidirectional
- Used for communication
  - Between processes on same system
  - Between processes on different machines (network)

Any process can act as:
- Client
- Server

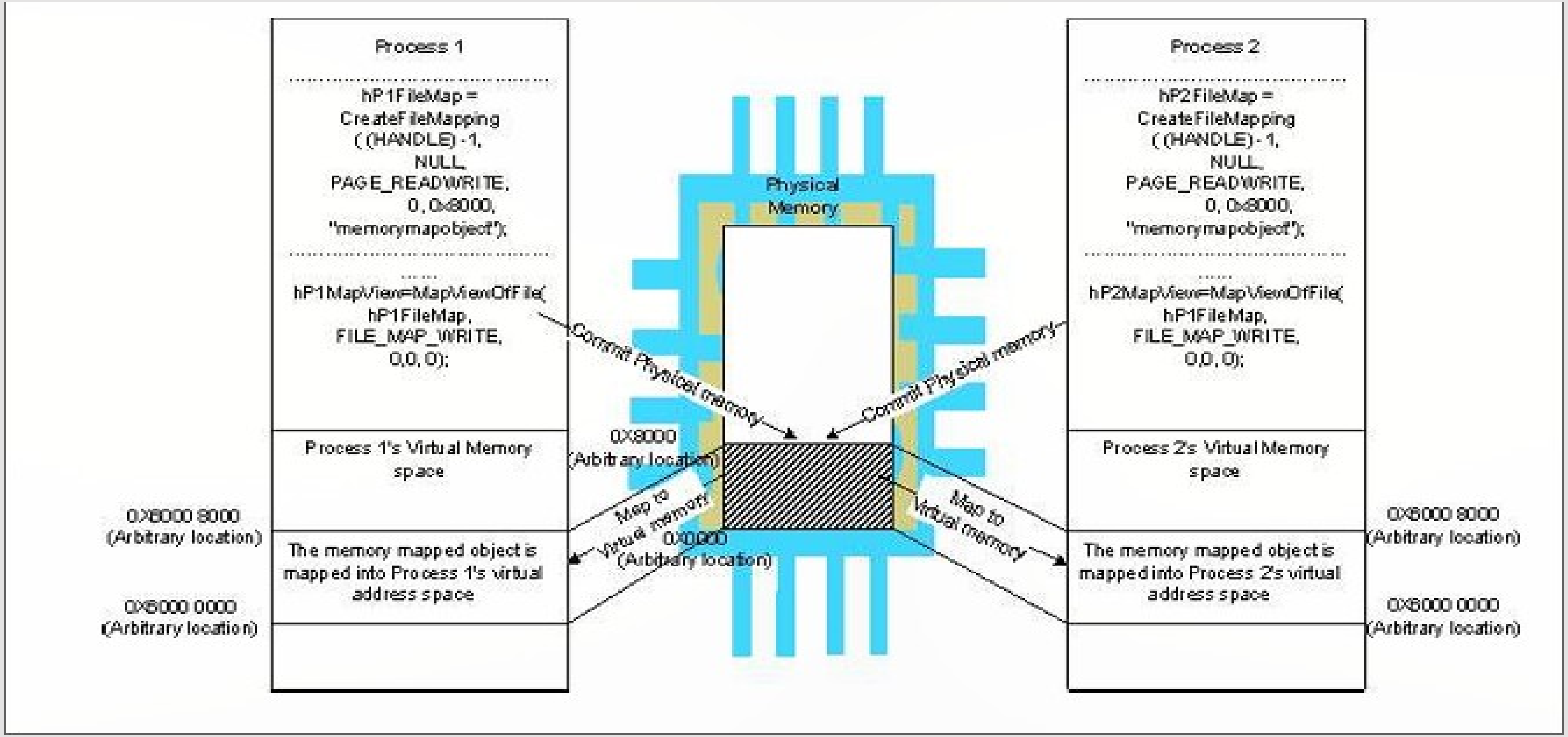
2. Memory Mapped Objects

A Memory Mapped Object is a shared memory technique used in some Real-Time Operating Systems (RTOS).

- It allocates a shared block of memory that can be accessed by multiple processes simultaneously

How it Works

- A mapping object is created
- Physical memory is reserved and committed
- A process maps this memory to its virtual address space


Process 1
hP1FileMap =
CreateFileMapping
((HANDLE) -1,
NULL,
PAGE_READWRITE,
0, 0x8000,
"memorymapobject");
......
hP1MapView=MapViewOfFile(
hP1FileMap,
FILE_MAP_WRITE,
0,0, 0);
Process 1's Virtual Memory space
0X6000 8000
(Arbitrary location)
The memory mapped object is mapped into Process 1's virtual address space
0X6000 0000
(Arbitrary location)
Physical Memory
Commit Physical memory
Commit Physical memory
0X8000
(Arbitrary location)
Map to Virtual memory
Map to Virtual memory
0X0000
(Arbitrary location)
Process 2
hP2FileMap =
CreateFileMapping
((HANDLE) - 1,
NULL,
PAGE_READWRITE,
0, 0x8000,
"memorymapobject");
......
hP2MapView=MapViewOfFile(
hP1FileMap,
FILE_MAP_WRITE,
0,0, 0);
Process 2's Virtual Memory space
0X6000 8000
(Arbitrary location)
The memory mapped object is mapped into Process 2's virtual address space
0X6000 0000
(Arbitrary location)

CONCLUSION

Shared memory is an efficient IPC mechanism that allows processes to communicate by accessing a common memory space. It provides high-speed data sharing with minimal overhead, but requires proper synchronization to avoid conflicts